

A LIKELIHOOD ESTIMATION OF HIV INCIDENCE INCORPORATING INFORMATION ON PAST PREVALENCE

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Summary: The prevalence and incidence of an epidemic are basic characteristics that are essential for study planning, assessing effect of interventions and for determining public health priorities. A direct approach for estimating incidence is to undertake a longitudinal cohort study where a representative sample of disease free individuals are followed for a specified period of time and new cases of infection are observed and recorded. This approach is expensive, time consuming and prone to bias due to loss-to-follow-up. An alternative approach is to estimate incidence from cross sectional surveys using biomarkers to identify persons recently infected as in Brookmeyer and Quinn (1995) and Janssen, Satten, Stramer, Rawal, O'Brien, Weiblen, Hecht, Jack, Cleghorn, Kahn, Chesney and Busch (1998). This paper builds on the work of Janssen et al. (1998) and extends the theoretical framework proposed by Balasubramanian and Lagakos (2010) by incorporating information on past prevalence and deriving maximum likelihood estimators of incidence. The performance of the proposed method is evaluated through a simulation study, and its use is illustrated using data from the Botswana AIDS Impact (BAIS) III survey of 2008.

1. Introduction

Knowledge of current human immunodeficiency virus (HIV) prevalence and incidence is essential for monitoring the impact of HIV, determining public health priorities, assessing the impact of interventions and for planning purposes (UNAIDS, 2006; Brookmeyer, Quinn, Shepherd, Mehendale, Rodrigues and Bollinger, 1995; Brookmeyer and Quinn, 1995; Janssen et al., 1998; Balasubramanian and Lagakos, 2010; Wang and Lagakos, 2009; Wang and Lagakos, 2010; Kaplan and Brookmeyer, 1999). However, obtaining accurate incidence estimates poses serious challenges. A direct approach is to estimate incidence from longitudinal cohort studies where a representative sample of disease free individuals are followed for a specified period of time and new cases of the disease are observed and recorded. Incidence rate is then computed as a ratio of the new cases of the disease detected over a given period to the number of person years (or other units of time used) of exposure (Kaplan and Brookmeyer, 1999). In poor resource settings, estimation of incidence from cohort studies is hampered by high cost and low follow up rates that may lead to follow up bias. This type of bias arises in settings where the incidence rates among the subgroups of individuals who return for follow-up are different from incidence rates among those who do not return for follow-up (Brookmeyer and Quinn, 1995; Brookmeyer et al., 1995). Other drawbacks of the longitudinal approach are that they can be time consuming and difficult to implement.

An alternative approach is to estimate HIV incidence from a cross sectional survey using biomarkers to identify persons acutely or recently infected (Brookmeyer and Quinn, 1995; Janssen et al., 1998). Both Brookmeyer and Quinn (1995) and Brookmeyer et al. (1995) proposed the detection of HIV-1 p24 antigen in the pre-seroconversion period for identifying persons with acute infection for estimating incidence. The idea is to determine the prevalence of p24 antigenemia among persons who have not seroconverted (Brookmeyer and Quinn, 1995). Incidence is estimated using standard epidemiological relationships between incidence, prevalence and mean duration in which prevalence is a product of incidence rate and mean duration (Freeman and Hutchison, 1980). However, this approach requires large sample sizes in order to obtain reliable estimates of incidence because the duration of antigenemia before antibody detection is short, about 22.5 days (Brookmeyer and Quinn, 1995; Janssen et al., 1998). Subsequently, another assay that is associated with a longer period from being undetectable to detectable was developed (Janssen et al., 1998). That is, the indirect immunoassay (EIA) is modified or “detuned” or made “less sensitive” by increasing the specimen dilution (Janssen et al., 1998; Parekh, Kennedy, Dobbs, Pau, Byers, Green, Hu, Vanichseni, Young, Choopanya, Mastro and McDougal, 2002). The detuned assay is then used to identify recent infection by detection of differential HIV-1 antibody titer. The assay have been used to estimate incidence in many US populations due to its availability in those settings (Janssen et al., 1998; Parekh et al., 2002). In general, identification of new infections is done in two stages. In the first stage, one takes a cross-sectional sample of size n from a population, and tests each person using a more sensitive test (typically ELISA). Individuals testing negative are assumed to be uninfected at that time. In the second stage, individuals testing positive on a more sensitive test are tested again using a less-sensitive test such as BED assays. Subjects that are found to be positive on a sensitive test but negative on the less sensitive test for HIV infection are considered to be recent infections. This method is known as the serologic testing algorithm for recent HIV seroconversion (STARHS), (Janssen et al., 1998). However, limited availability in other parts of the world and some other

limitations of the assay (that include requirements of special equipments, subtype dependent performance and significant variability of the window periods) hampered its wide usage and subsequently led to the evolution of IgG-capture BED-EIA (commonly referred to as BED assay) for detecting recent HIV-1 infections (Parekh et al., 2002; Parekh and McDougal, 2005; WHO, 2009). The BED assay has been used worldwide on five continents, both in general population and in high risk group (Barnighausen, McWalter, Rosner, Newell and Welte, 2010).

Subsequently Balasubramanian and Lagakos (2010) provides a statistical justification for the approach proposed by (Janssen et al., 1998) through the method of maximum likelihood estimation. (Balasubramanian and Lagakos, 2010) proposed a flexible statistical framework which allows extensions to settings where more than two diagnostic tests are used and incorporation of the covariates information such as age and gender.

In many settings, at the time of the cross sectional survey, the information on previous prevalences (existing/non-recent infections) from previous studies is often available. If the period between the immediate past survey and the current cross sectional survey is not too long, then one can use the prevalence from the past survey to improve incidence estimation in the current survey. Incorporating additional information of known past prevalence can lead to an efficiency gain in estimating incidence. This paper builds on the work of Janssen et al. (1998) and extends the theoretical framework proposed by (Balasubramanian and Lagakos, 2010) by incorporating information on past prevalence in order to improve the efficiency of incidence estimation.

It is known (McDougal, Parekh, Peterson, Branson, Dobbs, Ackers and Gurwith, 2006; Barin, Meyer, Lancar, Deveau, Gharib, Laporte, Desenclos and Costagliola, 2005; Chawla, Murphy, Donnelly, Booth, Johnson, Parry, Phillips and Geretti, 2007) that a varying proportion of individuals tested with BED assay produce the “false-recent” results long after seroconversion (assay non-progressors). That is, some long standing infection are misclassified as recent long after they have seroconverted (Karita, Price, Hunter, Chomba, Allen, Fei, Kamali, Sunders, Anzala, Katende, Ketter, the IAVI Collaborative Seroprevalence and Team, 2007; McDougal et al., 2006; Hargrove, Humphrey, Mutasa, Parekh, McDougal, Ntozinie, Chidawaniyika, Moulton, Ward, Nathoo, Ili and Kopp, 2008). Estimators that take into account false recent rate have been proposed (McWalter and Welte, 2010; Wang and Lagakos, 2009). Since our method is likelihood based, we extend the approach in Wang and Lagakos (2009) to make adjustments for false recent rate in the proposed model.

The rest of the paper is structured as follows. In Section 2, we review models for HIV progression and the likelihood estimators for incidence. The proposed model that incorporates the information on previous prevalence is introduced in Section 3. A simulation study to investigate the performance of this model is presented in Section 4. We use the data from Botswana AIDS Impact (BAIS) III survey of 2008 in Section 5 to illustrate the use of the proposed method. Lastly, we end the paper with a discussion in Section 6.

2. Models and the likelihood function

2.1. The 3-state model, state probabilities and incidence

We first consider a 3-state disease progression model for the natural history of HIV/AIDS denoted by S_1 , S_2 and S_3 as in Balasubramanian and Lagakos (2010). Here S_1 represents the “pre-

seroconversion” state which corresponds to the period in which an individual is either infected but have not yet seroconverted or is uninfected. S_2 represents the “recent infection” state where HIV antibodies are detectable by a sensitive diagnostic test but not yet through a less sensitive test. Finally S_3 represents the “non-recent infection” state in which HIV antibodies are detectable by a less sensitive test and sensitive test. We later consider the 4-state model to make adjustments for assay non-progressors.

We use similar notation as in Wang and Lagakos (2009), and Claggett, Lagakos and Wang (2012). Suppose T denotes the calendar time of HIV seroconversion for someone born at time 0. Let t denote the calendar time of the cross sectional sample. Furthermore, let $f(t)$, $F(t)$ and $\lambda(t)$ denote the incidence density function (incidence density) for becoming infected, cumulative distribution function and the incidence rate/hazard rate of T at time $t \geq 0$ respectively. Let L_2 denote the sojourn or residence time in S_2 with the corresponding cumulative distribution denoted by $G_2(\cdot)$. We assume that L_2 has support in $[0, L_2^*]$, where $L_2^* < t$ and is independent of T . It follows that $G_2(0) = 0$ and $G_2(L_2^*) = 1$.

The prevalence probabilities for the 3-states at time t are given by:

$$\begin{aligned}\pi_1(t) &\stackrel{def}{=} P(\text{in } S_1 \text{ at calendar time } t) \\ &= 1 - F(t)\end{aligned}\tag{1}$$

$$\begin{aligned}\pi_2(t) &\stackrel{def}{=} P(\text{in } S_2 \text{ at calendar time } t) \\ &= \int_{t-L_2^*}^t f(u) [1 - G_2(t-u)] du\end{aligned}\tag{2}$$

$$\pi_3(t) = 1 - \pi_1(t) - \pi_2(t)\tag{3}$$

Our interest is on the estimation of the HIV incidence rate at time t given by

$$\lambda(t) = \frac{f(t)}{1 - F(t)}.\tag{4}$$

Since t is the fixed calendar time at which the cross sectional sample is obtained, we shall hereafter denote $\lambda(t)$ by λ .

2.1.1. The Likelihood Function and HIV incidence estimation without previous prevalence

Suppose we select a random sample of size n from the population at some calendar time t . Let n_0 denote the number of subjects who test negative on a more sensitive test, n_r denote the number of subjects who test positive on a sensitive test and negative on a less sensitive test and n_1 denote the number of subjects who test positive on a less sensitive test such that $n = n_0 + n_r + n_1$. The n_r individuals are also referred to as “recent infections” while the n_1 individuals are referred to as “long standing infections” or “non-recent infections”. Note that if one tests negative on a sensitive test then a less sensitive test is generally not done but assumed to be negative.

The general likelihood function, L , corresponding to the 3-state model is given by

$$L = [\pi_1(t)]^{n_0} [\pi_2(t)]^{n_r} [\pi_3(t)]^{n_1}\tag{5}$$

subject to $\pi_1(t) + \pi_2(t) + \pi_3(t) = 1$.

It can be shown that the MLE of incidence rate is,

$$\hat{\lambda} = \frac{n_r}{n_0 \mu} \quad (6)$$

(Wang and Lagakos, 2009) and the variance of the estimate $\hat{\lambda}$ obtained from the observed Fisher information is given by

$$\text{var}(\hat{\lambda}) = \frac{n_r}{n_0^2 \mu^2}. \quad (7)$$

3. The extended model allowing for past prevalence

Suppose HIV prevalence at time $t_1 < t$, $F(t_1)$, is known. Thereafter we denote $F(t_1)$ by θ_0 . Let $\tau = t - t_1$ denote the time between the current study at time t and the immediate past study at time t_1 . Let the prevalence at time t be $F(t) = \theta$. Assuming constant density for HIV seroconversion in short period of time right before the cross sectional survey, that is,

$$f(u) = f \text{ for } u \in [t - L_2^*, t]$$

then θ is given by

$$\begin{aligned} \theta &= \theta_0 + \int_{t_1}^t f(u) du \\ &= \theta_0 + f(t - t_1) \text{ (for } t_1 < L_2^*) \\ &= \theta_0 + f\tau. \end{aligned} \quad (8)$$

Since

$$\lambda = \frac{f}{1 - \theta} = \frac{f}{1 - \theta_0 - f\tau}$$

then

$$f = \frac{\lambda(1 - \theta_0)}{1 + \lambda\tau}. \quad (9)$$

The likelihood function, in terms of λ , is given by:

$$L(\lambda) = \left[1 - \theta_0 - \frac{\lambda(1 - \theta_0)\tau}{1 + \lambda\tau} \right]^{n_0} \left[\frac{\lambda(1 - \theta_0)\mu}{1 + \lambda\tau} \right]^{n_r} \left[\theta_0 + \frac{\lambda(1 - \theta_0)(\tau - \mu)}{1 + \lambda\tau} \right]^{n_1}$$

Hence the maximum likelihood estimator of λ , $\hat{\lambda}$, when previous prevalence is known is given by

$$\hat{\lambda} = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad (10)$$

where

$$\begin{aligned} a &= n_0 \tau [\tau - (1 - \theta_0)\mu] \\ b &= (n_0 + n_1) \tau \theta_0 - (n_r + n_1) [\tau - (1 - \theta_0)\mu] \\ c &= -n_r \theta_0 \end{aligned}$$

for $b^2 - 4ac > 0$. Note that to ensure that the estimated incidence rate is a positive quantity we use the positive root of the quadratic in Eq (10). See the Appendix for a derivation of the quadratic equation.

We note that the proposed estimator, $\hat{\lambda}$, reduces to the simple estimator when we have no information on previous prevalence when τ equal μ , the mean window period. That is, for $\tau = \mu$,

$$\hat{\lambda} = \frac{n_r}{n_0\mu}.$$

This is the estimator of HIV incidence, given in Eq (6), when we have no previous prevalence.

To construct confidence intervals, we need the variance of $\hat{\lambda}$. Given $\frac{\partial^2 \ell}{\partial \lambda^2}$, the approximate variance of $\hat{\lambda}$,

$$\text{var}(\hat{\lambda}) = -E \left[\frac{\partial^2 \ell}{\partial \lambda^2} \right]^{-1}$$

where

$$\frac{\partial^2 \ell}{\partial \lambda^2} = \frac{n\tau^2}{(1 + \lambda\tau)^2} - \frac{n_r}{\lambda^2} - \frac{n_1 [\tau - (1 - \theta_0)\mu]^2}{[\theta_0 + \lambda \{ \tau - (1 - \theta_0)\mu \}]^2} \quad (11)$$

We obtain the estimate of the variance of $\hat{\lambda}$ by replacing λ in Eq (11) with $\hat{\lambda}$, its MLE given in Eq (10).

Also we note that for $\tau = \mu$

$$\text{var}(\hat{\lambda}) = \frac{n_r(n_0 + n_r)}{n_0^3\mu^2} \approx \frac{n_r}{n_0^2\mu^2}$$

as in Wang and Lagakos (2009). This is because $n_r \ll n_0$ hence $n_0 + n_r \approx n_0$.

3.1. Incorporating false recent rate into the extended model

The model we have introduced in Section 3 does not take into account individuals who remain negative on the less-sensitive assay indefinitely. It is recognized that a fraction of HIV infected subjects, referred to as assay non-reactors/non-progressors or elite suppressors, repeatedly test negative on a less sensitive test, such as BED assay, long after they have seroconverted leading in what has been termed “false recent rate” (Wang and Lagakos, 2009; McWalter and Welte, 2010; Brookmeyer, 2010; Hargrove et al., 2008; Novitsky, Wang, Kebaabetswe, Greenwald, Rossebkhane, Moyo, Musonda, Woldegabriel, Lagakos and Essex, 2009; Karita et al., 2007). Failure to account for this false recent rate results in estimates of incidence that are biased because they overestimate the number of subjects who are recently infected (Wang and Lagakos, 2009; Hargrove et al., 2008; Karita et al., 2007). Estimators to address this problem have been proposed (McWalter and Welte, 2010; Wang and Lagakos, 2009). Since the estimator we have introduced in Section 3 is also a maximum likelihood estimator, to incorporate the false recent rate we extend the approach of Wang and Lagakos (2009).

We consider the four-state progressive-disease model as in Wang and Lagakos (2009). Under this model, State 1 represents the “pre-seroconversion” state which corresponds to the period in which an individual is either infected but have not yet seroconverted or is uninfected. State 2 represents the “recent infection” state where HIV antibodies are detectable by a sensitive diagnostic test but not yet through a less sensitive test (though the test will eventually become reactive to the less-sensitive

assay). State 3 represents the “non-recent infection” state in which HIV antibodies are detectable by a less sensitive test and sensitive test. Finally state 4 represents subjects who are detectable by the sensitive test, but who will permanently remain nonreactive to the less-sensitive test.

3.2. The likelihood function and parameter estimation

Let p denote the proportion of subjects who will become reactive at some point after seroconversion. Hence the false recent rate, $1 - p$, denotes the proportion of subjects who remain negative on the less sensitive assay indefinitely. The log likelihood we present in Section 3 can be extended to incorporate the false recent rate as follows:

$$\begin{aligned}
 \ell(\lambda) &= n_0 \ln \left[1 - \theta_0 - \frac{\lambda(1 - \theta_0)\tau}{1 + \lambda\tau} \right] \\
 &\quad + n_r \ln \left[\frac{\lambda(1 - \theta_0)\mu p}{1 + \lambda\tau} + \left\{ \theta_0 + \frac{\lambda(1 - \theta_0)\tau}{(1 + \lambda\tau)} \right\} (1 - p) \right] \\
 &\quad + n_1 \ln \left[p \left\{ \theta_0 + \frac{\lambda(1 - \theta_0)(\tau - \mu)}{1 + \lambda\tau} \right\} \right] \\
 &\propto -n \ln(1 + \lambda\tau) + n_r \ln [\lambda\mu p - \lambda\mu p\theta_0 + \theta_0 - \theta_0 p + \lambda\tau - \lambda\tau p] \\
 &\quad + n_1 \ln [-\lambda\mu + \lambda\mu\theta_0 + \theta_0 + \lambda\tau].
 \end{aligned}$$

The partial derivative with respect to λ is

$$\frac{\partial \ell}{\partial \lambda} = -\frac{n\tau}{1 + \lambda\tau} + \frac{n_r m_r}{u_r + \lambda m_r} + \frac{n_1 m_1}{u_1 + \lambda m_1}, \quad (12)$$

where

$$\begin{aligned}
 n &= n_0 + n_r + n_1 \\
 m_r &= \mu p(1 - \theta_0) + \tau(1 - p) \\
 u_r &= \theta_0(1 - p) \\
 m_1 &= -\mu(1 - \theta_0) + \tau \\
 u_1 &= \theta_0.
 \end{aligned}$$

The MLE of $\hat{\lambda}$ is obtained by setting Eq (12) to 0 and is given by:

$$\hat{\lambda} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad (13)$$

where

$$\begin{aligned}
 a &= m_1 m_r \tau n_0 \\
 b &= m_r \tau u_1 (n_0 + n_1) + m_1 \tau u_r (n_0 + n_r) - m_1 m_r (n_r + n_1) \\
 c &= n \tau u_r u_1 - n_r m_r u_1 - n_1 m_1 u_r.
 \end{aligned}$$

We consider the positive root given by Eq (13) to ensure that the values of incidence rate are within acceptable range. Note that for this setting, values of incidence rate can be negative when $1 - p$ is unusually large say greater than 0.05.

The approximate variance of $\hat{\lambda}$, obtained by replacing λ by its maximum likelihood estimates given in Eq (13), is

$$\text{var}[\hat{\lambda}] = -\text{E} \left[\frac{\partial^2 \ell}{\partial \lambda^2} \right]^{-1},$$

where

$$\frac{\partial^2 \ell}{\partial \lambda^2} = \frac{n\tau^2}{(1 + \lambda\tau)^2} - \frac{n_r m_r^2}{(u_r + \lambda m_r)^2} - \frac{n_1 m_1^2}{(u_1 + \lambda m_1)^2} \quad (14)$$

and m_r, u_r, m_1 and u_1 are as defined before.

4. Simulation Study

We performed a simulation study to evaluate the performance of the proposed estimator of HIV incidence using 1000 simulations. We generated a sample of size $n = 3000$. The data was simulated from a multinomial distribution with parameters for the unadjusted model:

$$\begin{aligned} \pi_1(t) &= 1 - \theta_0 - \frac{\lambda(1 - \theta_0)\tau}{1 + \lambda\tau} \\ \pi_2(t) &= \frac{\lambda(1 - \theta_0)\mu}{1 + \lambda\tau} \\ \pi_3(t) &= 1 - \pi_1(t) - \pi_2(t). \end{aligned}$$

For the adjusted model that takes into account the false recent rate the parameters are:

$$\begin{aligned} \pi_1(t) &= 1 - \theta_0 - \frac{\lambda(1 - \theta_0)\tau}{1 + \lambda\tau} \\ \pi_2(t) &= \frac{\lambda(1 - \theta_0)\mu p}{1 + \lambda\tau} + \left\{ \theta_0 + \frac{\lambda(1 - \theta_0)\tau}{(1 + \lambda\tau)} \right\} (1 - p) \\ \pi_3(t) &= 1 - \pi_1(t) - \pi_2(t). \end{aligned}$$

Wang and Lagakos (2009) conduct an extensive simulation to evaluate the method that adjusts for false recent rate described in Section 3.1 in the absence of previous prevalence. We consider the proposed (unadjusted) model when we have previous prevalence (Model 2 denoted as M2) and compare it with the standard (unadjusted) model with no data on previous prevalence (Model 1 denoted as M1). We also consider the proposed (adjusted) model when we have previous prevalence (Model 4 denoted by M4) and compare it with the standard (adjusted) model (Model 3 denoted by M3) in order to determine in which settings previous prevalence improves the precision of incidence estimation.

For each simulation, we calculate the average point estimate of incidence based on the proposed model. The true value of the incidence rate, $\lambda(t)$ denoted by (I_{true}) is taken to be 3%. The average variance (standard errors) of these estimates are obtained from the observed Fisher information. We also obtain the empirical estimates of the variance from the 1000 simulated point estimate of

incidence. The results of the simulation study are presented in Table 1 for $\theta_0 = 0.10, 0.171, 0.2$. For each fixed value of θ_0 we assess how efficiency gain is affected for $\tau = 0.1, 0.3, 0.5, 0.6, 1.0$ years. Also presented are the values of the coverage probabilities. That is, the proportion of times out of 1000 simulations the 95% confidence interval covers the true value of HIV incidence. The results are presented for both unadjusted models (with and without previous prevalence).

We note that in all cases the simulated values of λ are closer to the true value of the incidence rate, $I_{true} = 0.03$. In general, $\hat{SE}(\hat{\lambda})$ and $E\hat{SE}(\hat{\lambda})$, the estimated standard errors and the corresponding empirical estimates, respectively, are very close to each other for all the different time points for both models. In most cases, the coverage probabilities are close to the nominal 95% confidence level. As depicted in Figure 1, suitable values of τ are those between the mean window period (where the two curves overlap) and one year. In particular, for the unadjusted model, there is an efficiency gain for values of τ between the mean window period and one year. In these settings we examined incorporating past prevalence leads to modest efficiency gain when past prevalence up to one year was used.

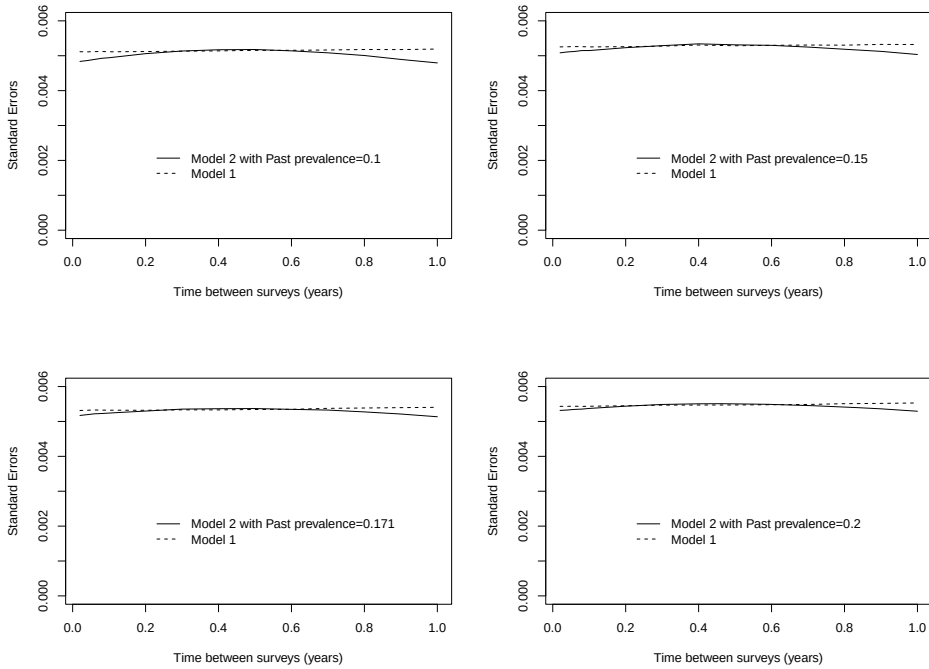


Figure 1: Graph of standard errors against time (years) between surveys using the data in Table 1

We also compared the adjusted models (with previous prevalence, $M4$ and without previous prevalence, $M3$). When we incorporate the false recent rate, we see that the estimates of λ are close to the true values in all cases and that the standard error increases for a higher proportion of

Table 1: Results of a simulation study comparing M2 and M1 with $N = 3000$ and $I_{true} = 0.03$. $\hat{\lambda}$, $\hat{SE}(\hat{\lambda})$ and $E\hat{SE}(\hat{\lambda})$ represents the average estimates, estimated standard errors and the corresponding empirical estimates respectively from 1000 simulations. 95%CV denotes the proportion of experiments in which I_{true} is contained in the nominal 95% confidence interval.

Simulation Results with $I_{true} = 0.03$							
τ (in years)	Estimates	Previous HIV Prevalence					
		$\theta_0 = 0.10$		$\theta_0 = 0.171$		$\theta_0 = 0.20$	
		M2	M1	M2	M1	M2	M1
0.1	$\hat{\lambda}$	0.0298	0.0299	0.0298	0.0298	0.0300	0.0300
	$\hat{SE}(\hat{\lambda})$	0.0049	0.0051	0.0052	0.0053	0.0054	0.0054
	$E\hat{SE}(\hat{\lambda})$	0.0051	0.0054	0.0053	0.0054	0.0053	0.0055
	95%CV	93.1	93.3	94.6	93.8	95.4	95.2
0.3	$\hat{\lambda}$	0.0299	0.0299	0.0298	0.0298	0.0302	0.0302
	$\hat{SE}(\hat{\lambda})$	0.0051	0.0051	0.0054	0.0053	0.0055	0.0055
	$E\hat{SE}(\hat{\lambda})$	0.0053	0.0054	0.0055	0.0055	0.0057	0.0057
	95%CV	92.8	92.7	93.7	93.8	94.0	94.1
0.5	$\hat{\lambda}$	0.0300	0.0300	0.0297	0.0297	0.0300	0.0300
	$\hat{SE}(\hat{\lambda})$	0.0052	0.0052	0.0054	0.0053	0.0055	0.0055
	$E\hat{SE}(\hat{\lambda})$	0.0053	0.0053	0.0056	0.0056	0.0055	0.0055
	95%CV	93.5	92.8	93.5	93.3	94.7	94.3
0.6	$\hat{\lambda}$	0.0296	0.0299	0.0297	0.0297	0.0300	0.0300
	$\hat{SE}(\hat{\lambda})$	0.0051	0.0051	0.0053	0.0053	0.0055	0.0055
	$E\hat{SE}(\hat{\lambda})$	0.0054	0.0054	0.0057	0.0058	0.0056	0.0057
	95%CV	92.7	92.4	92.7	92.9	93.9	93.8
1	$\hat{\lambda}$	0.0300	0.0300	0.0299	0.0300	0.0302	0.0302
	$\hat{SE}(\hat{\lambda})$	0.0048	0.0052	0.0051	0.0054	0.0053	0.0055
	$E\hat{SE}(\hat{\lambda})$	0.0050	0.0056	0.0052	0.0056	0.0054	0.0057
	95%CV	93.7	94.0	94.9	94.2	94.5	94.5

false recent rate increases (see Table 2). Wang and Lagakos (2009) observed similar results. The estimated standard errors are closer to the corresponding empirical estimates in both cases of false recent proportions namely 5% and 2%. The efficiency gain in the proposed adjusted model compared to the standard adjusted model (Wang and Lagakos, 2009) remains modest.

Table 2: Results of a simulation study with the incorporation of the False Recent Rate (FRR) denoted by p for $N = 3000$ and $I_{true} = 0.03$. $\hat{\lambda}$, $\hat{SE}(\hat{\lambda})$ and $E\hat{SE}(\hat{\lambda})$ represents the average estimates, estimated standard errors and the corresponding empirical estimates respectively from 1000 simulations. 95%CV denotes the proportion of experiments in which I_{true} is contained in the nominal 95% confidence interval.

Simulation Results with $I_{true} = 0.03$ incorporating FRR									
$\tau(y)$	Est	$\theta_0 = 0.10$				$\theta_0 = 0.20$			
		$p = 0.95$		$p = 0.98$		$p = 0.95$		$p = 0.98$	
		M4	M3	M4	M3	M4	M3	M4	M3
0.3	$\hat{\lambda}$	0.0306	0.0306	0.0304	0.0304	0.0304	0.0303	0.0304	0.0304
	$\hat{SE}(\hat{\lambda})$	0.0063	0.0063	0.0056	0.0057	0.0078	0.0079	0.0065	0.0065
	$E\hat{SE}(\hat{\lambda})$	0.0064	0.0064	0.0058	0.0058	0.0080	0.0080	0.0068	0.0068
	95%CV	94.5	94.8	94.5	94.2	94.8	94.2	93.1	93.2
0.5	$\hat{\lambda}$	0.0304	0.0304	0.0304	0.0304	0.0304	0.0303	0.0303	0.0302
	$\hat{SE}(\hat{\lambda})$	0.0061	0.0064	0.0056	0.0057	0.0076	0.0079	0.0065	0.0066
	$E\hat{SE}(\hat{\lambda})$	0.0064	0.0066	0.0059	0.0060	0.0080	0.0082	0.0068	0.0069
	95%CV	94.3	93.9	93.6	93.9	93.8	94.1	93.7	93.2
0.6	$\hat{\lambda}$	0.0305	0.0305	0.0303	0.0303	0.0305	0.0305	0.303	0.0304
	$\hat{SE}(\hat{\lambda})$	0.0061	0.0064	0.0056	0.0057	0.0076	0.0080	0.0064	0.0066
	$E\hat{SE}(\hat{\lambda})$	0.0063	0.0067	0.0059	0.0060	0.0079	0.0083	0.0068	0.0069
	95%CV	94.7	94.6	93.9	93.7	94.2	93.6	93.6	93.5
1	$\hat{\lambda}$	0.0306	0.0307	0.0303	0.0303	0.0305	0.0307	0.0305	0.0305
	$\hat{SE}(\hat{\lambda})$	0.0054	0.0066	0.0051	0.0058	0.0068	0.0082	0.0060	0.0067
	$E\hat{SE}(\hat{\lambda})$	0.0054	0.0057	0.0053	0.0060	0.0071	0.0084	0.0063	0.0069
	95%CV	94.5	95.6	94.0	94.0	93.7	93.3	93.7	94.0

5. Illustration of the proposed model using BAIS III data set

5.1. Introduction

We applied the proposed model to the Botswana AIDS impact survey (BAIS III) data of 2008. BAIS III was the third sexual behaviour national population level survey. More importantly, BAIS

III has gone beyond the traditional aims of assessing knowledge, attitude and behavior regarding HIV and AIDS to estimating the levels of HIV prevalence and incidence in the general population aged 18 months and above (CSO, 2008). This was an important exercise particularly with regard to assessment of the impact of national programs and it also provided the basis of future research including this current paper. BAIS III was a cross sectional study.

5.2. Data and Results

A parallel testing algorithm using commercial ELISA test kits (Vironostika-HIV Uni-Form II plus O (Organon Teknika, Boxtel, The Netherlands) and Murex (Abbott, Wiesbaden, Germany)) were used to screen blood samples. A specimen was considered HIV antibody positive if it was reactive on parallel ELISA testing otherwise it was considered HIV antibody negative. In order to detect recent HIV-1 seroconversion, all the HIV positive specimens were tested again using a less sensitive test, Aware BED enzyme immunoassay (EIA) HIV-1 incidence Test (Calypse Biomedical Corporation, Portland, Oregon, USA). Specimens were considered recent HIV-1 seroconversion if they were not reactive to Aware BED EIA test (tested negative) for $ODn < 0.8$ otherwise the specimens were classified as long standing infections (for more details of these Laboratory Testing, see CSO (2008)).

All persons aged 18 months and above were eligible for HIV testing using a sample of size $n = 21,414$ during the survey period. But only 67% (sample size=14,351) provided blood specimen for HIV Testing. Of the 14,351 subjects who tested, 2521 tested HIV positive, $n_0 = 11,823$ tested HIV negative, while 7 results were indeterminate. Out of the 2521 subjects who tested positive for HIV, there were $n_r = 149$ recent infections and $n_1 = 2319$ long standing infections. For the remaining subjects, $n_{others} = 53$ specimens were reported missing (for 27 specimens the box was not found and for the other 26 the blood samples were finished). We assumed that the missing specimens were missing completely at random. That is, the events that lead to any particular data-item being missing are independent both of observable variables and of unobservable parameters of interest (Little and Rubin, 2002).

Estimated HIV incidence rate per 100 person-years (I) was calculated using the four estimators:

- a) Model 1 (M1): The standard unadjusted estimator (Kaplan and Brookmeyer, 1999; Balasubramanian and Lagakos, 2010)
- b) Model 2 (M2): The proposed unadjusted estimator incorporating past prevalence
- c) Model 3 (M3): The standard adjusted estimator (Wang and Lagakos, 2009)
- d) Model 4 (M4): The proposed adjusted estimator incorporating past prevalence

As cautioned by Hargrove et al. (2008), McWalter and Welte (2010), and Wang and Lagakos (2009), the use of locally available estimated parameters cannot be overemphasized. Because the locally estimated parameters are not available the values used in this paper are for illustration purposes only. However, the comparison and conclusions still remain valid with these assumed parameter values. In both scenarios of Model 1 versus Model 2 and Model 3 versus Model 4 we assume a mean window period of 155 days (CSO, 2008). We use a value of $\tau = 1$ year. For models 3 and 4, we assume a false recent rate, $1 - p = 1.5\%$. In Botswana, between 2004 and 2008, overall national

HIV prevalence increased from 17.1% to 17.6% (CSO, 2008) for individuals aged 1.5 years and above. For purposes of illustration we used simple extrapolation method to estimate the previous prevalence for 2007. The values of θ_0 are provided in Table 3.

Table 3: Values of θ_0

Demographics		Sample n (n_0, n_r, n_1)	$\theta_0 \approx \frac{n_1 + n_r}{n} - 0.00125$
Sex	Male	6516 (5603, 55, 858)	0.139
	Female	7775 (6220, 94, 1461)	0.199
Age (y)	≤ 4	872 (854, 5, 13)	0.019
	5-14	3382 (3242, 18, 122)	0.040
	15-19	1449 (1395, 5, 49)	0.036
	20-29	2967 (2401, 43, 523)	0.190
	30-39	2142 (1287, 33, 822)	0.398
	40-49	1429 (928, 27, 474)	0.349
	50+	2050 (1716, 18, 316)	0.162
Overall	All groups	14291 (11823, 149, 2319)	0.171

The 95% confidence intervals (95% CI) are provided for all the 4 models. We also use all the four models to estimate group specific HIV incidence by sex and age. The results for all the four models are presented in Table 4.

In both cases, Model 2 (compared to Model 1) and Model 4 (compared to Model 3) produce better estimates of precision as shown by the narrow confidence intervals. These are settings where we have assumed the previous prevalence is known. We can see that, in all the four models, incidence is higher for females compared to males as it was reported in the BAIS III report. Also, incidence is higher for age groups 20-49 years compared to other age groups. There is a substantial decrease in the estimated incidence when the assumed false recent rate is 1.5%. It is therefore important to make necessary adjustments where there is need because failure to do that will result in overestimation of incidence while making unnecessary adjustments will lead to underestimation of incidence.

6. Discussion

We propose a method for estimating incidence of a disease, such as HIV, from a cross-sectional study when previous prevalence is known. We find that in some settings such a modification can modestly improve the precision of the estimator of incidence. In order to compare our model (which assumes known previous prevalence) with the standard model (which assumes no previous prevalence) we assume a known value of the mean window period of 155 days, the same as that used in BIAS III data analysis (CSO, 2008). Although uncertainty in the false recent rate and the mean window period affects estimators of incidence, we note that for comparison between methods the two uncertainties will not affect the comparative conclusions. The models (with and without previous prevalence) were compared under same conditions of not accounting for uncertainty in both the false recent

Table 4: Estimated HIV incidence by gender and age using all the four Models

Estimated HIV incidence in Botswana					
Demographics		$I_{M1}(95\% \text{ CI})$	$I_{M2}(95\% \text{ CI})$	$I_{M3}(95\% \text{ CI})$	$I_{M4}(95\% \text{ CI})$
Sex	Male	2.31 (1.70, 2.92)	1.97 (1.46, 2.48)	1.76 (1.14, 2.38)	1.45 (0.94, 1.96)
	Female	3.56 (2.84, 4.28)	3.00 (2.41, 3.59)	2.72 (2.00, 3.44)	2.19 (1.59, 2.79)
Age (y)	1-4	1.38 (0.17, 2.59)	0.89 (0.14, 1.64)	1.32 (0.10, 2.54)	0.83 (0.08, 1.58)
	5-14	1.31 (0.71, 1.91)	1.02 (0.57, 1.47)	1.17 (0.56, 1.78)	0.89 (0.44, 1.34)
	15-19	0.84 (0.10, 1.58)	0.70 (0.11, 1.29)	0.72 (0.00, 1.47)	0.58 (0.00, 1.18)
	20-29	4.22 (2.96, 5.48)	3.43 (2.43, 4.43)	3.44 (2.17, 4.71)	2.68 (1.67, 3.69)
	30-39	6.04 (3.98, 8.10)	5.20 (3.46, 6.94)	3.75 (1.67, 5.83)	3.06 (1.20, 4.82)
	40-49	6.85 (4.27, 9.43)	5.69 (3.59, 7.79)	5.02 (2.42, 7.62)	3.94 (1.82, 6.06)
	50+	2.47 (1.33, 3.61)	2.12 (1.17, 3.07)	1.81 (0.66, 2.96)	1.50 (0.53, 2.47)
Overall	All groups	2.97 (2.49, 3.45)	2.51 (2.12, 2.90)	2.26 (1.78, 2.74)	1.84 (1.44, 2.24)

rate and assumed mean window period. Methods of handling for these uncertainties have been proposed by Wang and Lagakos (2010) through the idea of augmented designs where all or some individuals who are tested as recently infected are followed until they exit the recent infection state. The design allows for internal estimation of both the false recent rate and the mean window period and subsequently take into account the uncertainties associated with these estimators. To ensure that our model did not overestimate incidence as a result of assay non-progression, we extend the idea of (McWalter and Welte, 2010; Wang and Lagakos, 2009) through a likelihood approach to account for false recent rate.

It is important to note that in the era of improved standard of care where most people living with HIV receive antiretroviral therapy (ARV) which may suppress the viral load below detection limit, it is possible that some individuals on ARV may potentially ‘return’ or ‘revert’ back to the recent infection state (Wang and Lagakos, 2009; Brookmeyer, 2010; McWalter and Welte, 2010). However it has been noted that where ARV usage is documented, all subjects on treatment can be assumed to be in non-recent state because treatment initiation is often long after sero-conversion; usually after 1 to 2 years depending on the genetic structure of the individual (Wang and Lagakos, 2009) which can slow or enhance disease progression.

Alternatively, to overcome this problem, a novel though expensive approach is to design an assay, with high sensitivity and specificity, for identifying new infections based on the characteristics of HIV gene diversification within an infected individual (Park, Love, Nelson, Thurston, Perelson and Lee, 2011).

One of the findings of the current research is that incidence estimates differ between males and females and between age groups. Hence, future research should aim to incorporate covariates dependence on incidence rate estimation. An additional future enhancement to the proposed method will also be to incorporate uncertainty in past prevalence.

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References

- BALASUBRAMANIAN, R. AND LAGAKOS, S. W. (2010). Estimating HIV incidence based on combined prevalence testing. *Biometrics*, **66**, 1–10. DOI: 10.1111/j.1541-0420.2009.01242.x.
- BARIN, F., MEYER, L., LANCAR, R., DEVEAU, C., GHARIB, M., LAPORTE, A., DESENCLOS, J., AND COSTAGLIOLA, D. (2005). Development and validation of an immunoassay for identification of recent human immunodeficiency virus type 1 infections and its use on dried serum spots. *Journal of Clinical Microbiology*, **43** (9), 4441–4447. Doi:10.1128/JCM.43.9.4441-4447.2005.
- BARNIGHAUSEN, T. A., MCWALTER, T. A., ROSNER, Z., NEWELL, M. L., AND WELTE, A. (2010). HIV incidence estimation using the BED capture enzyme immunoassay: systematic review and sensitivity analysis. *Epidemiology*, **21**(5), 685–697.
- BROOKMEYER, R. (2010). On the statistical accuracy of biomarker assays for HIV incidence. *Journal of Acquired Immune Deficiency Syndrome*, **54** (4), 406–414.
- BROOKMEYER, R. AND QUINN, T. C. (1995). Estimation of current Human Immunodeficiency Virus incidence rates from a cross-sectional survey using early diagnostic tests. *American Journal of Epidemiology*, **141**, 166–172.
- BROOKMEYER, R., QUINN, T. C., SHEPHERD, M., MEHENDALE, S., RODRIGUES, J., AND BOLLINGER, R. (1995). The AIDS epidemic in India: a method for estimating current human immunodeficiency virus (HIV) incidence rates. *American Journal of Epidemiology*, **142** (7), 709–713.
- CHAWLA, A., MURPHY, G., DONNELLY, C., BOOTH, C. L., JOHNSON, M., PARRY, J. V., PHILLIPS, A., AND GERETTI, A. M. (2007). Human immunodeficiency virus (HIV) antibody avidity testing to identify recent infection in newly diagnosed HIV type 1 (HIV-1)-seropositive persons infected with diverse HIV-1 subtypes. *Journal of Clinical Microbiology*, **45** (2), 415–420. Doi:10.1128/JCM.01879-06.
- CLAGGETT, B., LAGAKOS, S. W., AND WANG, R. (2012). Augmented Cross-Sectional Studies with Abbreviated Follow-up for Estimating HIV Incidence. *Biometrics*, **68** (1), 62–74. DOI: 10.1111/j.1541-0420.2011.01632.x.
- CSO (2008). BOTSWANA AIDS IMPACT SURVEY III. Technical report, Botswana Government.
- FREEMAN, J. AND HUTCHISON, G. B. (1980). Prevalence, incidence and duration. *American Journal of Epidemiology*, **112** (5), 707–723.

- HARGROVE, J. W., HUMPHREY, J. H., MUTASA, K., PAREKH, B. S., MCDUGAL, J. S., NTOZINIE, R., CHIDAWANIYIKA, H., MOULTON, L. H., WARD, B., NATHOO, K., ILI, P. J., AND KOPP, E. (2008). Improved HIV-1 incidence estimates using the BED capture enzyme immunoassay. *AIDS*, **22**, 511–518.
- JANSSEN, R. S., SATTEN, G. A., STRAMER, S. L., RAWAL, B. D., O'BRIEN, T. R., WEIBLEN, B. J., HECHT, F. M., JACK, N., CLEGHORN, J., KAHN, J. O., CHESNEY, M. A., AND BUSCH, M. P. (1998). New testing strategy to detect early HIV-1 infection for use in incidence estimates and for clinical and prevention purposes. *Journal of the American Medical Association*, **280**, 42–48.
- KAPLAN, E. H. AND BROOKMEYER, R. (1999). Snapshot estimators of recent HIV incidence rates. *Operations Research*, **47** (1), 29–37.
- KARITA, E., PRICE, M., HUNTER, E., CHOMBA, E., ALLEN, S., FEI, L., KAMALI, A., SUNDERS, E. J., ANZALA, O., KATENDE, M., KETTER, N., THE IAVI COLLABORATIVE SEROPREVALENCE, AND TEAM, I. S. (2007). Investigating the utility of the HIV-1 BED capture enzyme immunoassay using cross sectional and longitudinal seroconverter specimens from Africa. *AIDS*, **21**, 403–408.
- LITTLE, R. J. A. AND RUBIN, D. B. (2002). *Statistical analysis with missing data*. Second edition. John Wiley and Sons.
- MCDUGAL, J. S., PAREKH, B. S., PETERSON, M. L., BRANSON, B. M., DOBBS, T., ACKERS, M., AND GURWITH, M. (2006). Comparison of HIV type I incidence observed during longitudinal follow-up with incidence estimated by cross-sectional analysis using the BED capture enzyme immunoassay. *AIDS Research and Human Retroviruses*, **22**, 945–952.
- MCWALTER, T. A. AND WELTE, A. (2010). Relating recent infection prevalence to incidence with a sub-population of assay non-progressors. *Journal of Mathematical Biology*, **60**, 687–710. DOI: 10.1007/s00285-009-0282-7.
- NOVITSKY, V., WANG, R., KEBABETSWE, L., GREENWALD, J., ROSSEBKHAN, R., MOYO, S., MUSONDA, R., WOLDEGABRIEL, E., LAGAKOS, S. W., AND ESSEX, M. (2009). Better control of early viral replication is associated with slower rate of elicited antibodies in the Detuned EIA during primary HIV-1C infection. *Journal of Acquired Immune Deficiency Syndrome*, **52**(2), 265–272.
- PAREKH, B. S., KENNEDY, M. S., DOBBS, T., PAU, C.-P., BYERS, R., GREEN, T., HU, D. J., VANICHSENI, S., YOUNG, N. L., CHOOPANYA, K., MASTRO, T. D., AND MCDUGAL, J. S. (2002). Quantitative detection of increasing HIV type 1 antibodies after seroconversion: A simple assay for detecting recent HIV infection and estimating incidence. *AIDS Research and Human Retroviruses*, **18** (4), 295–307.
- PAREKH, B. S. AND MCDUGAL, J. S. (2005). Application of laboratory methods for estimation of HIV-1 incidence. *Indian Journal of Medical Research*, **121**, 510–518.
- PARK, S. Y., LOVE, T. M. T., NELSON, J., THURSTON, S. W., PERELSON, A. S., AND LEE, H. Y. (2011). Designing a Genome-Based HIV Incidence Assay with High Sensitivity and Specificity. *AIDS*, **25**(16), F13–F19. DOI: 10.1097/QAD.0b013e328349f089.
- UNAIDS (2006). Report on the global AIDS epidemic. Available: http://www.unaids.org/en/HIV_data/2006GlobalReport/default.asp. Geneva: United Nations. Accessed 8 March 2008.
- WANG, R. AND LAGAKOS, S. W. (2009). On the use of adjusted cross-sectional estimators of HIV incidence. *Journal of Acquired Immune Deficiency Syndrome*, **52**(5), 538–547.

WANG, R. AND LAGAKOS, S. W. (2010). Augmented cross-sectional prevalence testing for estimating HIV incidence. *Biometrics*, **66** (3), 864–874. DOI: 10.1111/j.1541-0420.2009.01356.x.
 WHO (2009). WHO Technical Working Group on HIV Incidence Assays. Available: http://www.who.int/diagnostics_laboratory/links/hiviwg_capetown_07_09.pdf. Accessed 28 May 2012.

Appendix: Estimating Quadratic Equation

The log likelihood function, $\ell(\lambda)$, where now only λ is unknown, for the unadjusted model where the previous prevalence is known, can be simplified to:

$$\begin{aligned}\ell(\lambda) = & -n_0 \log(1 + \lambda \tau) + k + n_r \log(\lambda) - n_r \log(1 + \lambda \tau) \\ & + n_1 \log[\theta_0 + \lambda(\tau - \mu + \theta_0 \mu)] - n_1 \log(1 + \lambda \tau)\end{aligned}$$

Hence we can write,

$$\ell(\lambda) = -n \log(1 + \lambda \tau) + k + n_r \log(\lambda) + n_1 \log[\theta_0 + \lambda \{\tau - (1 - \theta_0)\mu\}]$$

Where

$$k = n_0 \log(1 - \theta_0) + n_r \log[(1 - \theta_0)\mu]$$

and

$$n = n_0 + n_r + n_1.$$

Thus

$$\frac{\partial \ell}{\partial \lambda} = -\frac{n\tau}{1 + \lambda \tau} + \frac{n_r}{\lambda} + \frac{n_1 [\tau - (1 - \theta_0)\mu]}{\theta_0 + \lambda [\tau - (1 - \theta_0)\mu]}$$

and by setting $\frac{\partial \ell}{\partial \lambda} = 0$ leads to

$$\frac{n\tau}{1 + \lambda \tau} = \frac{n_r}{\lambda} + \frac{n_1 [\tau - (1 - \theta_0)\mu]}{\theta_0 + \lambda [\tau - (1 - \theta_0)\mu]}$$

Simplifying the above equation, we obtain a quadratic equation in λ :

$$\lambda^2 n_0 \tau [\tau - (1 - \theta_0)\mu] + \lambda [n\tau\theta_0 - n_r\theta_0\tau - (n_r + n_1) \{\tau - (1 - \theta_0)\mu\}] - n_r\theta_0 = 0$$

